

Michael Zhan

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Professional Experience

Bloomberg | Software Engineer *New York, NY*

Jan 2025 – Present

- Designed and implemented an attribute-based authorization engine using AST-based policy representation and partial evaluation, enabling extensible, fine-grained access control for millions of entities.
- Engineered the Go backend of a content management platform spanning APIs, data modeling, storage, authorization, and content delivery; implemented access-controlled tree projections over cyclic, multi-parent collection graphs for hierarchical content retrieval.
- Led a large architectural refactor of a six-year-old Go CMS, replacing disparate entity-specific implementations with standardized interfaces and shared services, eliminating ~30K lines of duplicated and boilerplate code.
- Built a data pipeline that transformed a heterogeneous graph of CMS content into a normalized, AI-compatible schema, deriving content-specific metadata such as hierarchical URLs to power natural-language retrieval on two Bloomberg websites serving tens of thousands of users monthly.

Amazon | Software Development Engineer Intern *Austin, TX*

May 2024 – Aug 2024

- Optimized internal AWS service onboarding using Java, Ruby, and Spring dependency injection by extracting a microservice from a monolithic package, reducing onboarding time by 90%.
- Deployed to production in all supported regions, benefiting 10,000 internal AWS customers.
- Received full-time offer following internship.

Georgiou Research Group, UT Austin | Undergraduate Research Assistant *Austin, TX*

Jan 2020 – May 2023

- Developed and implemented single-linkage hierarchical clustering algorithms for >100,000 antibody sequence strings in Python and C++. Ran 50% faster than the previous internal tool, with identical results.
- Built k-means clustering workflow with Python scikit-learn and R to discover new gene segments in ferrets using outlier detection. Discovered 2 new genes with this method.

Genentech, Inc. | Software Engineer Intern *San Francisco, CA*

Jan 2022 – Dec 2022

- Developed modular automation package for ML dataset generation and visualization with Python, reducing processing time by over 80%. Tool is currently in use across multiple Genentech sites.
- Engineered soft sensor machine learning models for cell culture parameter prediction using scikit-learn and AutoML libraries with an emphasis on feature optimization. Attained R^2 values of over 0.95.

Personal Projects

Stock Data Streaming System

Jul 2026

- Built a real-time stock data pipeline in Go, simulating prices with geometric Brownian motion and streaming data to a frontend via Server-Sent Events (SSE).
- Implemented an $O(1)$ -amortized sliding-window max algorithm to track rolling price ranges in real time and visualized streaming data using ChartGPU.

Mock Log System

Jun 2026

- Built a distributed log-processing pipeline in C++ using Kafka, TimescaleDB, and PgBouncer, decoupling log production and consumption for asynchronous ingestion.
- Implemented custom HTTP handling in C++ to enable frontend queries and time-series analytics.

Custom Language Interpreter

Nov 2025

- Implemented lexing, Pratt parsing, and AST evaluation in Go for a custom language with primitives, functions, arrays, and hash maps. Structured interpreter for modularity and extensibility.

Technical Skills

- Languages:** Go, Python, C++, C, TypeScript, JavaScript, Java
- Developer Tools:** Docker, Git, AWS, PostgreSQL, Kafka, CI/CD, Linux, GCP, Jira

Education

Harvard University

Sep 2023 – Dec 2024

M.S. Computational Science and Engineering (GPA: 3.96)

The University of Texas at Austin, Engineering Honors Program

Sep 2019 – May 2023

B.S. Chemical Engineering, Elements of Computing Certificate (GPA: 3.99)